

A candidate full classification of two-block q -balls as λ -intersection bodies

Automated mathematical discovery run `fa_banach_001`
Lane 8, agent `agent_lane_08`, model GPT5.6

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Status. Candidate full solution, likely valid, needs expert review. The main review target is the regular Weber–Schafheitlin transform in Lemma 2. Novelty was checked only by a bounded search.

Abstract

Boris Rubin asked whether the two-block q -ball $B_{q,\ell}^n$, for $q > 2$, is a λ -intersection body in the interval $\max(n - \ell, \ell) - 2 < \lambda < n - 3$. We give an affirmative answer and, combined with Rubin’s negative result, obtain the classification

$$B_{q,\ell}^n \in \mathcal{I}_\lambda^n \iff \lambda \geq \max(n - \ell, \ell) - 2, \quad 0 < \lambda < n.$$

The proof writes the inverse norm power as a positive mixture of anisotropic copies of a product-ball inverse norm. A Weber–Schafheitlin formula and Euler’s hypergeometric transformation prove positivity of the Fourier transform of the product kernel. The threshold case follows by closure of positive tempered distributions.

1 Source question and exact scope

The source is Boris Rubin, *Intersection Bodies and Generalized Cosine Transforms*. It is arXiv:0704.0061v2, Section 7, printed page 34 [1]. For $1 \leq \ell < n$ and $q > 0$, Rubin writes

$$B_{q,\ell}^n = \left\{ x = (x', x'') \in \mathbb{R}^{n-\ell} \oplus \mathbb{R}^\ell : |x'|^q + |x''|^q \leq 1 \right\}.$$

His open problem is:

Let $q > 2$, $\ell > 1$. Is $B_{q,\ell}^n$ a λ -intersection body if

$$\max(n - \ell, \ell) - 2 < \lambda < n - 3?$$

Rubin’s Theorem 5.1 says, for $0 < \lambda < n$, that a star body K lies in $\mathcal{I}_\lambda^n$ if and only if $\|\cdot\|_K^{-\lambda}$ is a positive-definite tempered distribution. Rubin’s Proposition 7.5 proves non-membership when $0 < \lambda < \max(n - \ell, \ell) - 2$ [1]. Thus only the positive direction and the endpoint need to be established.

body for any $0 < \lambda < n$.

Proof. Owing to Lemma 7.1, the function (7.8) (with p replaced by $-\lambda$) is positive, and therefore, by Lemma 7.3, $\|\cdot\|_{q,\ell}^{-\lambda}$ represents a positive definite distribution. Now the result follows by Theorem 5.1. $\square$

Consider the case $q > 2$. In this case $B_{q,\ell}^n$ is convex, and, owing to Example 6.2, $B_{q,\ell}^n \in \mathcal{I}_\lambda^n$ for all $n-3 \leq \lambda < n$. What about $\lambda < n-3$? This case is especially intriguing.

Proposition 7.5. *If $q > 2$ and $0 < \lambda < \max(n-\ell, \ell) - 2$, then $\|\cdot\|_{q,\ell}^{-\lambda}$ is not a positive definite distribution and therefore, $B_{q,\ell}^n \notin \mathcal{I}_\lambda^n$.*

Proof. Let $0 < \lambda < n - \ell - 2$ and suppose the contrary, that $B_{q,\ell}^n \in \mathcal{I}_\lambda^n$. Consider the section of $B_{q,\ell}^n$ by the $(n - \ell + 1)$ -dimensional plane $\eta = \mathbb{R}e_n \oplus \mathbb{R}^{n-\ell}$. By Theorem 5.12, $B_{q,\ell}^n \cap \eta \in \mathcal{I}_\lambda^{n-\ell+1}$ in η , and therefore

$$\|x_n e_n + x''\|_{q,\ell}^\lambda = (|x_n|^q + |x''|^q)^{-\lambda/q}$$

is a positive definite distribution in η . By the second derivative test (see [K4, Theorem 4.19]) this is impossible if $0 < \lambda < n - \ell - 2$. A similar contradiction can be obtained if we assume $0 < \lambda < \ell - 2$ and consider the section of $B_{q,\ell}^n$ by the $(\ell + 1)$ -dimensional plane $\mathbb{R}e_1 \oplus \mathbb{R}^\ell$. $\square$

Proposition 7.5 can be proved without using the second derivative test and Theorem 5.12 on sections of λ -intersection bodies; see [R4]. The bounds for λ appear to be the same.

Open problem. *Let $q > 2$, $\ell > 1$. Is $B_{q,\ell}^n$ a λ -intersection body if $\max(n - \ell, \ell) - 2 < \lambda < n - 3$?*

This problem does not occur in the case $\ell = 1$ as in Example 6.6.

Figure 1: Rubin's Proposition 7.5 and open problem, rendered from printed page 34 of the source PDF.

2 Proof intuition

Put $a = n - \ell$ and $b = \ell$. The first observation is an exact identity: the function $(|x'|^q + |x''|^q)^{-\lambda/q}$ is a probability mixture of

$$\max(|x'|/c, |x''|)^{-\lambda}, \quad c > 0.$$

These are inverse norm powers of linear images of the Cartesian product $B_2^a \times B_2^b$. It is therefore enough to understand one product body.

The Fourier transform of $\max(|x|, |y|)^{-\lambda}$ is obtained by a layer-cake formula. Fourier transforms of Euclidean balls introduce two Bessel functions, and their remaining radial integral is a Weber–Schafheitlin integral. In the chamber $|\xi| > |\eta|$ the answer is a positive constant times

$$\frac{|\xi|^{\lambda-n}}{\Gamma((\lambda+2-b)/2)} {}_2F_1\left(\frac{n-\lambda}{2}, \frac{b-\lambda}{2}; \frac{b}{2} + 1; \frac{|\eta|^2}{|\xi|^2}\right).$$

Although the second upper parameter need not be positive, Euler's transformation replaces the two upper parameters by $(\lambda+2-a)/2$ and $(\lambda+2)/2$. They, and the Gamma denominator, are positive exactly when $\lambda > \max(a, b) - 2$. The other chamber is symmetric. The possible singularity on $|\xi| = |\eta|$ is locally integrable in this range, so the Fourier transform is a nonnegative function. Positive definiteness follows, positive mixtures preserve it, and the endpoint follows by distributional closure.

3 Two lemmas

Lemma 1 (Exact product mixture). *For $q > 0$, $\lambda > 0$, and $r, s > 0$,*

$$(r^q + s^q)^{-\lambda/q} = \int_0^\infty \max(r/c, s)^{-\lambda} p_{q,\lambda}(c) dc, \quad (1)$$

where

$$p_{q,\lambda}(c) = (q + \lambda)c^{q-1}(1 + c^q)^{-\lambda/q-2}. \quad (2)$$

Moreover, $p_{q,\lambda}$ is a probability density on $(0, \infty)$.

Proof. The substitution $u = c^q$ gives

$$\int_0^\infty p_{q,\lambda}(c) dc = \frac{q + \lambda}{q} \int_0^\infty (1 + u)^{-\lambda/q-2} du = 1.$$

Set $t = r/s$ and

$$M(t) = \int_0^t c^\lambda p_{q,\lambda}(c) dc.$$

Direct differentiation gives

$$M(t) = t^{\lambda+q}(1 + t^q)^{-\lambda/q-1}.$$

Splitting the right side of (1) at $c = t$ and dividing by $s^{-\lambda}$ gives

$$t^{-\lambda}M(t) + \int_t^\infty p_{q,\lambda}(c) dc.$$

Its derivative is $-\lambda t^{-\lambda-1}M(t)$, the same as the derivative of $(1 + t^q)^{-\lambda/q}$, and both expressions tend to 1 as $t \downarrow 0$. This proves (1). $\square$

Lemma 2 (The Euclidean product kernel). *Let $a, b \geq 1$, $n = a + b$, and $0 < \lambda < n$. If*

$$\lambda > \max(a, b) - 2, \quad (3)$$

then the tempered distribution

$$F_{a,b,\lambda}(x, y) = \max(|x|, |y|)^{-\lambda}, \quad (x, y) \in \mathbb{R}^a \oplus \mathbb{R}^b,$$

is positive definite.

Proof. Use the Fourier convention $\widehat{f}(\zeta) = \int_{\mathbb{R}^n} f(u) e^{-iu \cdot \zeta} du$. The layer-cake formula is an identity in $\mathcal{S}'(\mathbb{R}^n)$:

$$F_{a,b,\lambda}(x, y) = \lambda \int_0^\infty t^{-\lambda-1} \mathbf{1}_{tB_2^a}(x) \mathbf{1}_{tB_2^b}(y) dt. \quad (4)$$

Indeed, it is pointwise true away from the origin; the small- t end is controlled by t^n , and the large- t end by the rapid decay of the test function.

Write $A = |\xi|$, $B = |\eta|$. The standard ball transform

$$\widehat{\mathbf{1}_{tB_2^d}}(\xi) = (2\pi)^{d/2} t^{d/2} |\xi|^{-d/2} J_{d/2}(t|\xi|)$$

turns (4), away from the coordinate axes, into a positive constant times

$$A^{-a/2} B^{-b/2} \int_0^\infty t^{n/2-\lambda-1} J_{a/2}(At) J_{b/2}(Bt) dt. \quad (5)$$

Let $\rho = n/2 - \lambda - 1$. For $A > B > 0$, the Weber–Schafheitlin formula, interpreted by Abel regularization and analytic continuation in the parameters, gives

$$\begin{aligned} \widehat{F}_{a,b,\lambda}(\xi, \eta) &= C_{a,b,\lambda} \frac{A^{\lambda-n}}{\Gamma((\lambda+2-b)/2)} \\ &\times {}_2F_1\left(\frac{n-\lambda}{2}, \frac{b-\lambda}{2}; \frac{b}{2} + 1; \frac{B^2}{A^2}\right), \end{aligned} \quad (6)$$

where $C_{a,b,\lambda} > 0$. This is the classical chamber formula when the integral is ordinary and its regular-distribution continuation in the remaining cases. Wrochna gives the corresponding arbitrary-exponent distributional formula [2]. Here (3) implies

$$\rho = \frac{n}{2} - \lambda - 1 < 1.$$

Thus the continuation in the radial ratio is a regular, locally integrable distribution; in particular, there is no additional delta-type term on $A = B$. Concretely, the large-argument Bessel asymptotics reduce the only non-smooth transverse term to an Abel limit of

$$\int_1^\infty t^{\rho-1} \cos((A-B)t + \theta) dt.$$

For $0 < \rho < 1$ this is a locally integrable multiple of $|A-B|^{-\rho}$ (with side-dependent phases), for $\rho = 0$ it is logarithmic, and for $\rho < 0$ it is continuous. Delta derivatives arise only after the exponent reaches the nonregular regime $\rho \geq 1$. Formula (6) also follows by inserting $\mu = a/2$, $\nu = b/2$, $\rho = n/2 - \lambda - 1$ in the usual Weber–Schafheitlin parameters. The Gamma denominator then becomes exactly $\Gamma((\lambda+2-b)/2)$.

Set $z = B^2/A^2 \in (0, 1)$. Euler's transformation gives

$$\begin{aligned} {}_2F_1\left(\frac{n-\lambda}{2}, \frac{b-\lambda}{2}; \frac{b}{2} + 1; z\right) \\ = (1-z)^{\lambda+1-n/2} {}_2F_1\left(\frac{\lambda+2-a}{2}, \frac{\lambda+2}{2}; \frac{b}{2} + 1; z\right). \end{aligned} \quad (7)$$

All coefficients of the hypergeometric series on the right are positive, because (3) implies $(\lambda+2-a)/2 > 0$. The Gamma denominator in (6) is positive because $(\lambda+2-b)/2 > 0$. Hence (6) is positive.

When $B > A$, the symmetric calculation interchanges a and b and is again positive. The axes are obtained as nonnegative limits of the chamber formula away from the origin. At the frequency origin the transform is homogeneous of degree $\lambda - n$, which is locally integrable because $\lambda > 0$. Near $A = B > 0$, the worst possible singularity in (7) is a logarithm or a power

$$|1-z|^{\lambda+1-n/2}.$$

It is locally integrable since

$$\lambda > \max(a, b) - 2 \geq n/2 - 2 \implies \lambda + 1 - n/2 > -1.$$

Consequently $\widehat{F}_{a,b,\lambda}$ is represented on all of $\mathbb{R}^n$ by a nonnegative locally integrable function. Therefore $F_{a,b,\lambda}$ is positive definite. $\square$

Remark 1 (Why the exponent restriction matters). The Weber–Schafheitlin exponent is $\rho = n/2 - \lambda - 1$. At $\lambda > \max(a, b) - 2$ it is strictly below 1, which is precisely the regular-distribution side of the cone singularity. At a symmetric endpoint $a = b$ one can have $\rho = 1$; the proof below deliberately obtains that endpoint by closure instead of asserting the function formula there.

4 Full classification

Theorem 1. *Let $1 \leq \ell < n$, $q > 2$, and $0 < \lambda < n$. Then*

$$B_{q,\ell}^n \in \mathcal{I}_\lambda^n \iff \lambda \geq \max(n - \ell, \ell) - 2. \quad (8)$$

When the right-hand threshold is nonpositive, it imposes no restriction beyond $\lambda > 0$.

Proof. Put $a = n - \ell$, $b = \ell$, and first assume $\lambda > \max(a, b) - 2$. Lemma 1, with $r = |x'|$ and $s = |x''|$, yields

$$\|(x', x'')\|_{q,\ell}^{-\lambda} = \int_0^\infty \max(|x'|/c, |x''|)^{-\lambda} p_{q,\lambda}(c) dc. \quad (9)$$

The integrand is the inverse λ -power of the Minkowski functional of the anisotropic product $cB_2^a \times B_2^b$. It is a nonsingular linear image of the product in Lemma 2, so it is positive definite. Linear changes of variables preserve positive definiteness. Equation (9) is also valid in $\mathcal{S}'$: both sides are locally integrable for $\lambda < n$, and the positive integral can be paired with Schwartz test functions by the same small/large scale bounds used in (4). Thus the right side is a positive mixture of positive-definite distributions. It is positive definite, and Rubin's Theorem 5.1 gives $B_{q,\ell}^n \in \mathcal{I}_\lambda^n$.

Let $\lambda_0 = \max(a, b) - 2$. If $\lambda_0 > 0$, choose $\lambda_j \downarrow \lambda_0$ with $\lambda_j > \lambda_0$. Equivalence of the product norm and the Euclidean norm, followed by dominated convergence near zero and rapid decay at infinity, gives

$$\|\cdot\|_{q,\ell}^{-\lambda_j} \longrightarrow \|\cdot\|_{q,\ell}^{-\lambda_0} \quad \text{in } \mathcal{S}'(\mathbb{R}^n).$$

The cone of positive-definite tempered distributions is closed, so the limit is positive definite. This proves membership at the threshold. If $\lambda_0 \leq 0$, every allowed $\lambda > 0$ already lies in the strict range.

Conversely, Rubin's Proposition 7.5 states that for $q > 2$ and $0 < \lambda < \max(a, b) - 2$, the distribution $\|\cdot\|_{q,\ell}^{-\lambda}$ is not positive definite and $B_{q,\ell}^n \notin \mathcal{I}_\lambda^n$ [1]. This completes the classification. $\square$

Corollary 1 (Answer to the source problem). *For every parameter choice in Rubin's interval*

$$q > 2, \quad \ell > 1, \quad \max(n - \ell, \ell) - 2 < \lambda < n - 3,$$

the body $B_{q,\ell}^n$ is a λ -intersection body.

5 Verification and limitations

Analytic checks

The following points were checked independently while preparing the packet.

- The density in (2) has total mass one, and its truncated λ -moment differentiates to the claimed closed form.
- Substitution into the Weber–Schafheitlin formula gives upper parameters $(n - \lambda)/2$ and $(b - \lambda)/2$, and Gamma denominator $\Gamma((\lambda + 2 - b)/2)$ in the chamber $A > B$.
- Euler's transformation gives exactly (7).
- The cone exponent is greater than -1 , and the origin homogeneity is locally integrable.
- The endpoint is not inferred from the chamber formula; it is handled by closedness in $\mathcal{S}'$.

Computational sanity check

The reusable script `code/verifier.py` checks normalization and (1) on 36 parameter tuples, and checks Euler’s transformation and positivity on 30 dimension/ratio tuples, including ratios approaching the cone. From the repository root, run

```
conda run --no-capture-output -n sandbox python
runs/fa_banach_001/solutions/full/
0704.0061_q_l_ball_intersection_classification/code/verifier.py
```

These finite evaluations are not part of the proof.

Main human-review focus

An expert should first verify the transition from (5) to (6), including the Fourier normalization (irrelevant to sign), the parameter continuation, and the assertion that $\rho < 1$ leaves a regular locally integrable distribution rather than an additional term supported on $A = B$. The included Wrochna PDF is the supporting arbitrary-exponent reference. The positive mixture and endpoint closure are elementary but should also be audited.

Novelty search

A bounded search on 22 July 2026 used the run’s registry, solution, attempt, and proof-gap indexes; exact searches for arXiv:0704.0061, $B_{q,\ell}^n$, “ (q, ℓ) -ball”, “lambda-intersection body”, and “ $\max(n - \ell, \ell) - 2$ ”; citation-title inspection for the source DOI; and Rubin’s related arXiv:math/0701317. Searches combining Cartesian products of Euclidean balls with intersection bodies and Weber–Schafheitlin transforms also found no explicit later solution. This search was not exhaustive and does not establish originality.

6 References

References

- [1] B. Rubin, *Intersection Bodies and Generalized Cosine Transforms*, arXiv:0704.0061v2 (2007); see Theorem 5.1, Proposition 7.5, and the open problem on printed p. 34. <https://arxiv.org/abs/0704.0061>.
- [2] M. Wrochna, *Weber–Schafheitlin Integrals with Arbitrary Exponent*, arXiv:1004.5518 (2010). <https://arxiv.org/abs/1004.5518>.